

Gabriel Ong

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Summary / Statement

I have a strong interest in the intersection of Artificial Intelligence and Synthetic Biology, as well as of Biology and Chemistry, and intend to pursue a research career in computational protein design. I hope to be able to work closely with both academia and industry to address real-world problems in sustainability, for instance through novel enzymes for green chemistry. I have rich, holistic research experience in diverse fields, and strong skills in integrating knowledge and concepts across various disciplines to gain novel perspectives for problem solving and discovery. I am also a recipient of the highly competitive and prestigious A*STAR National Science Scholarship, demonstrating my passion and aptitude for scientific research.

Education

The Scripps Research Institute, San Diego, California, United States

PhD Student, Computational Biology and Modeling, Aug 2025 to 2030 (expc.)

University of Cambridge, Cambridgeshire, United Kingdom

Undergraduate Student, Natural Sciences (Biological), Sep 2020 to Jun 2024

Year 1 modules: Biology of Cells, Chemistry, Evolution & Behaviour, Mathematical Biology

Mathematical Biology was accompanied by a programming class with R

Ranked 39 out of 627 in Year 1 (top 6%)

Year 2 modules: Biochemistry & Molecular Biology (BMB), Chemistry B (Organic & Inorganic), Pharmacology

Each module was accompanied by a laboratory class with at least 60 hours of practical experience

Biochemistry skills acquired: Cloning, PCR, Western blot, ELISA, protein purification

Chemistry skills acquired: Organic and inorganic synthesis, IR and NMR spectroscopy

Pharmacology skills acquired: Tissue drug response assays (guinea pig ileum), antimicrobial assays

Ranked 70 out of 575 (Class I) in Year 2 (top 12%). Ranked 3rd & 4th in Biochemistry and Pharmacology respectively

Year 3 specialisation: Biochemistry

Project title: Bioinformatics studies of polyketide synthase dehydratase and enoyl reductase domains

Skills: Python, Pymol, protein-protein docking, AlphaFold

Ranked 17 out of 44 (Class II.1) in Year 3 (top 38%)

Year 4 Integrated Masters: Systems Biology

Project title: Protein structure networks for variant effect prediction

Skills: Python and Machine Learning

Ranked 5 out of 20 (Class I) in Year 4 (top 25%)

Won Best Research Project award

Hwa Chong Institution, Singapore

GCE 'A' Level, 7 distinctions, 2012 to 2017

Editorial team for school science publication Voyager 2016

Top in Junior College 1 cohort for Chemistry

Academic Employment

A*STAR Institute of Sustainability for Chemicals, Energy and Environment, Singapore

- Research Engineer, Jul 2024 - Jul 2025
- Protein design using protein language models (PLMs)
- Performed structural analysis and molecular dynamics simulations of a PLM-designed unstable RNase A for biomedical applications to better understand mutations suggested by the PLM. Manuscript in review
- Also trained models to predict chemical structures from biosynthetic gene cluster sequences, and vice versa, with the aim of expanding the toolkit for retrosynthesis to include enzymes and metabolic pathways for sustainable chemical production. Manuscript in review
- Developing in silico screening workflow for PLM-guided enzyme engineering for biomedical and sustainability applications
- Mentored by Dr Winston Koh and Dr Lim Yee Hwee

European Bioinformatics Institute, Cambridgeshire, United Kingdom

- Research Student, Oct 2023 - Jun 2024
- Bioinformatics project on protein structure networks for machine learning-guided variant effect prediction
- Used network-based features derived from network representations of protein structure, in combination with physicochemical properties, to train ML models using deep mutational scanning data for predicting of mutational effects on proteins. Manuscript in preparation
- Gained experience with Python and Machine Learning
- Presented at Frontiers in Science Research Summit at London in 2024
- Won Best Research Project Award
- Mentored by Dr Thodoris Koutsandreas and Dr Evangelia Petsalaki

Department of Biochemistry, University of Cambridge, Cambridgeshire, United Kingdom

- Research Student, Jan 2023 - Mar 2023
- Bioinformatics studies of polyketide synthase dehydratase and enoyl reductase domains
- Used computational protein-protein docking to identify key residues involved in enzyme-substrate and inter-enzyme interactions, with the aim of guiding polyketide synthase engineering efforts for sustainable natural product biosynthesis
- Gained experience with Python, PyMOL, molecular docking, and Linux
- Mentored by Dr Bill Broadhurst

European Bioinformatics Institute, Cambridgeshire, United Kingdom

- Summer Research Student, Jun 2022 - Aug 2022
- Bioinformatics project on enzyme evolution, with emphasis on catalytic residues and mechanism
- Created computational pipeline to infer evolutionary relationships between enzymes, integrating multimodal information on sequence, structure, active site conformation, and reaction mechanisms. This pipeline was adapted to for a study investigating enzyme convergent evolution, which has been published in FEBS Journal.
- Gained experience with Python, PyMOL, Linux, Snakemake, software pipeline development, and various EBI software tools
- Mentored by Prof Dame Janet Thornton

A*STAR Singapore Institute of Food and Biotechnology Innovation (SIFBI), Singapore

- Summer Research Student, Jul 2021 - Sep 2021
- Isolation of bioactive natural product compounds from Fungus and Actinomycetes
- Gained experience with Natural Product Chemistry, HPLC, NMR, Mass Spectrometry
- Mentored by Dr Yoganathan Kanagasundaram

A*STAR Bioinformatics Institute (BII), Singapore

- Research Intern, Feb 2020 - Aug 2020
- Bioinformatics project involving receptor-ligand docking relevant to cancer research and drug design
- Benchmarking of various protein-ligand docking software for rigid and flexible docking using KRAS as a model system
- Gained experience with PyMOL, Python, Linux, and molecular docking software
- Mentored by Dr Srinivasaraghavan Kannan and Dr Chandra Verma

Nanyang Environment and Water Research Institute (NEWRI), Singapore

- Research Intern, Jun 2016 - Dec 2016
- Optimisation of aquaporin-based biomimetic membranes for water treatment through reverse osmosis
- Won Silver Award and Institute of Engineers Special Award at Singapore Science and Engineering Fair 2017
- Won Merit and Best Team Award at Green Wave Environmental Care Competition 2016
- Mentored by Dr Qi Saren and Prof Wang Rong

Publications

Unified Genomic and Chemical Representations Enable Bidirectional Bio-synthetic Gene Cluster and Natural Product Retrieval

Guimei Liu, Yiting Li, **Gabriel Ong**, Fong Tian Wong, Dillon W.P. Tay, Yee Hwee Lim, Chuan Sheng Foo, Winston Koh (2025)
bioRxiv; doi: <https://doi.org/10.1101/2025.05.31.656985>

Paradigms of convergent evolution in enzymes

Ioannis G. Riziotis, Jenny C. Kafas, **Gabriel Ong**, Neera Borkakoti, Antonio J.M. Ribeiro, Janet M. Thornton (2024)
FEBS Journal; doi: <https://doi.org/10.1111/febs.17332>

Non-academic Employment

Singapore Armed Forces, Singapore

Automotive Technician (National Service), Jan 2018 - Jan 2020

Corporal First Class (CFC) Rank

Forward Maintenance Section for Exercise Panzerstrike in Germany during Autumn 2019

Worked as part of a team to meet tight deadlines, and adapted to changing requirements in a fast-paced work environment

Awards

Best Poster, Bioinformatics Institute (A*STAR) Poster Session, 2024

Best Research Project Award, Part III Systems Biology, University of Cambridge, 2024

A*STAR National Science Scholarship (BS-PhD), 2019

Silver Award and Special Award (Institute of Engineers), Singapore Science and Engineering Fair, 2017

Merit and Best Team Award, Green Wave Environmental Care Competition, 2016

Academic Service

Cambridge University Synthetic Biology Society

Webmaster, 2021-2022

Non-academic Service

Cambridge University Travel Society

Secretary, 2023-2024

Webmaster, 2022-2023

Churchill May Ball 2022 Committee

Webmaster

Language

Fluent in English and Mandarin Chinese.

August 30, 2025